

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : MANDATE PLUS

Other means of identification : Not applicable

Recommended use : Sanitizer

Product dilution information : 0,13 % - 0,26 %

Company : Ecolab, S. DE R.L DE C.V.
Industrial Avenue #28
Industrial Fractionation, Cuamatla, Cuautitlan Izcalli, Mexico State
Mexico 54730
(0155) 5864-1800

Quimiproduktos S. de R.L. de C.V.
Carretera Miguel Alemán, 990
Colonia La Encarnación - Apodaca, Nuevo León Mexico Zip Code:
66633
(0181) 47395900

Emergency telephone : 01 800 837 2998 Please request assistance in Spanish language

Issuing date : 29.08.2017

SECTION 2. HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Oxidizing liquids, Category 3	H272
Corrosive to Metals, Category 1	H290
Acute toxicity (Oral), Category 4	H302
Acute toxicity (Inhalation), Category 4	H332
Acute toxicity (Dermal), Category 3	H311
Skin corrosion, Category 1	H314
Serious eye damage, Category 1	H318
Acute aquatic toxicity, Category 2	H401

Product AT USE DILUTION

Acute aquatic toxicity, Category 2	H401
------------------------------------	------

GHS label elements
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements	: H272	May intensify fire; oxidizer.
	H290	May be corrosive to metals.
	H302 + H332	Harmful if swallowed or if inhaled.
	H311	Toxic in contact with skin.
	H314	Causes severe skin burns and eye damage.
	H401	Toxic to aquatic life.

SAFETY DATA SHEET

MANDATE PLUS

Precautionary Statements

: **Prevention:**

- P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
- P220 Keep away from clothing and other combustible materials.
- P234 Keep only in original container.
- P261 Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.
- P264 Wash skin thoroughly after handling.
- P270 Do not eat, drink or smoke when using this product.
- P271 Use only outdoors or in a well-ventilated area.
- P273 Avoid release to the environment.
- P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

- P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell. Rinse mouth.
- P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.
- P304 + P340 + P310 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/ physician.
- P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.
- P361 + P364 Take off immediately all contaminated clothing and wash it before reuse.
- P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
- P390 Absorb spillage to prevent material damage.

Storage:

- P405 Store locked up.

Disposal:

- P501 Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Hazard Statements

- : H401 Toxic to aquatic life.

Precautionary Statements

: **Prevention:**

- P273 Avoid release to the environment.

Disposal:

- P501 Dispose of contents/ container to an approved waste disposal plant.

Product AS SOLD

Other hazards

- : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

SAFETY DATA SHEET

MANDATE PLUS

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Acetic acid	64-19-7	10 - 20
nitric acid	7697-37-2	10 - 20
	7664-38-2	5 - 10
Secondary Alkanesulphonates	5324-84-5	5 - 10
Nonanoic acid	112-05-0	5 - 10
Decanoic Acid	334-48-5	1 - 5

Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention.
- Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

- In case of eye contact : Rinse with plenty of water.
- In case of skin contact : Rinse with plenty of water.
- If swallowed : Rinse mouth. Get medical attention if symptoms occur.
- If inhaled : Get medical attention if symptoms occur.

SECTION 5. FIRE-FIGHTING MEASURES

Product AS SOLD

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.
- Specific hazards during fire : Oxidizer. Contact with other material may cause fire.

SAFETY DATA SHEET

MANDATE PLUS

fighting

- Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
Nitrogen oxides (NO_x)
Sulfur oxides
Oxides of phosphorus
- Special protective equipment for fire-fighters : Use personal protective equipment.
- Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

- Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.
- Conditions for safe storage : Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from strong bases. Keep away from combustible

SAFETY DATA SHEET

MANDATE PLUS

material. Keep out of reach of children. Keep container tightly closed.
Store in suitable labeled containers.

Storage temperature : -10 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
Acetic acid	64-19-7	LMPE-CT	15 ppm 37 mg/m3	MX OEL
		LMPE-PPT	10 ppm 25 mg/m3	MX OEL
		VLE-PPT	10 ppm	NOM-010-STPS-2014
		VLE-CT	15 ppm	NOM-010-STPS-2014
Acetic acid	64-19-7	TWA	10 ppm	ACGIH
		STEL	15 ppm	ACGIH
		STEL	15 ppm 37 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	OSHA Z1
Nitric acid	7697-37-2	LMPE-PPT	2 ppm 5 mg/m3	MX OEL
		LMPE-CT	4 ppm 10 mg/m3	MX OEL
		VLE-PPT	2 ppm	NOM-010-STPS-2014
		VLE-CT	4 ppm	NOM-010-STPS-2014
nitric acid	7697-37-2	TWA	2 ppm	ACGIH
		STEL	4 ppm	ACGIH
		STEL	4 ppm 10 mg/m3	NIOSH REL
		TWA	2 ppm 5 mg/m3	NIOSH REL
		TWA	2 ppm 5 mg/m3	OSHA Z1
Phosphoric acid	7664-38-2	LMPE-PPT	1 mg/m3	MX OEL
		LMPE-CT	3 mg/m3	MX OEL
		VLE-PPT	1 mg/m3	NOM-010-STPS-2014
		VLE-CT	3 mg/m3	NOM-010-STPS-2014
	7664-38-2	TWA	1 mg/m3	ACGIH
		STEL	3 mg/m3	ACGIH

SAFETY DATA SHEET

MANDATE PLUS

		TWA	1 mg/m3	NIOSH REL
		STEL	3 mg/m3	NIOSH REL
		TWA	1 mg/m3	OSHA Z1
Acetic acid	64-19-7	LMPE-CT	15 ppm 37 mg/m3	MX OEL
		LMPE-PPT	10 ppm 25 mg/m3	MX OEL
		VLE-PPT	10 ppm	NOM-010-STPS-2014
		VLE-CT	15 ppm	NOM-010-STPS-2014
Acetic acid	64-19-7	TWA	10 ppm	ACGIH
		STEL	15 ppm	ACGIH
		STEL	15 ppm 37 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	NIOSH REL
		TWA	10 ppm 25 mg/m3	OSHA Z1
Nitric acid	7697-37-2	LMPE-PPT	2 ppm 5 mg/m3	MX OEL
		LMPE-CT	4 ppm 10 mg/m3	MX OEL
		VLE-PPT	2 ppm	NOM-010-STPS-2014
		VLE-CT	4 ppm	NOM-010-STPS-2014
nitric acid	7697-37-2	TWA	2 ppm	ACGIH
		STEL	4 ppm	ACGIH
		STEL	4 ppm 10 mg/m3	NIOSH REL
		TWA	2 ppm 5 mg/m3	NIOSH REL
		TWA	2 ppm 5 mg/m3	OSHA Z1
Phosphoric acid	7664-38-2	LMPE-PPT	1 mg/m3	MX OEL
		LMPE-CT	3 mg/m3	MX OEL
		VLE-PPT	1 mg/m3	NOM-010-STPS-2014
		VLE-CT	3 mg/m3	NOM-010-STPS-2014
	7664-38-2	TWA	1 mg/m3	ACGIH
		STEL	3 mg/m3	ACGIH
		TWA	1 mg/m3	NIOSH REL
		STEL	3 mg/m3	NIOSH REL
		TWA	1 mg/m3	OSHA Z1

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of

SAFETY DATA SHEET

MANDATE PLUS

degradation or chemical breakthrough.

- Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing
- Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

- Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

- Eye protection : No special protective equipment required.
- Hand protection : No special protective equipment required.
- Skin protection : No special protective equipment required.
- Respiratory protection : No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Color	: light yellow	colorless
Odor	: pungent	pungent
pH	: 1,0, (100 %)	2,5 - 3,5
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: No data available	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 1,17	
Water solubility	: No data available	
Solubility in other solvents	: No data available	
Partition coefficient: n-	: No data available	

SAFETY DATA SHEET

MANDATE PLUS

octanol/water

Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available
Explosive properties	: No data available
Oxidizing properties	: No data available
Molecular weight	: No data available
VOC	: No data available

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: None known.
Incompatible materials	: Bases Metals
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NO _x) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Toxic in contact with skin. Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.

SAFETY DATA SHEET

MANDATE PLUS

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Pain, Corrosion
Skin contact : Redness, Pain, Corrosion
Ingestion : Corrosion, Abdominal pain
Inhalation : Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact : No symptoms known or expected.
Skin contact : No symptoms known or expected.
Ingestion : No symptoms known or expected.
Inhalation : No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity : Acute toxicity estimate : > 300 mg/kg
Acute inhalation toxicity : 4 h Acute toxicity estimate : > 1 mg/l
Acute dermal toxicity : Acute toxicity estimate : > 200 mg/kg
Skin corrosion/irritation : No data available
Serious eye damage/eye irritation : No data available
Respiratory or skin sensitization : No data available
Carcinogenicity : No data available
Reproductive effects : No data available
Germ cell mutagenicity : No data available
Teratogenicity : No data available
STOT-single exposure : No data available
STOT-repeated exposure : No data available
Aspiration toxicity : No data available

SECTION 12. ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects : Toxic to aquatic life.

Product

Toxicity to fish : 96 h LC50: > 1 mg/l

SAFETY DATA SHEET

MANDATE PLUS

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Ingredients

Toxicity to daphnia and other aquatic invertebrates : Acetic acid
48 h EC50 Daphnia magna (Water flea): 39,6 mg/l

48 h EC50 Daphnia magna (Water flea): > 100 mg/l

Secondary Alkanesulphonates
48 h EC50 Daphnia: 3.200 mg/l

Nonanoic acid
48 h EC50 Daphnia: 96 mg/l

Ingredients

Toxicity to algae : Acetic acid
72 h EC50 Skeletonema costatum (marine diatom): 1.000 mg/l

72 h EC50 Desmodesmus subspicatus (green algae): > 100 mg/l

Persistence and degradability

Product AS SOLD

Readily biodegradable.

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : The product should not be allowed to enter drains, water courses or

SAFETY DATA SHEET

MANDATE PLUS

Disposal considerations	the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
	: Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport: generally applicable for transport in Mexico.

UN number	: 1760
Description of the goods	: CORROSIVE LIQUID, N.O.S. (acetic acid, Nitric acid)
Class	: 8
Packing group	: II
Environmentally hazardous	: no

Sea transport (IMDG/IMO)

UN number	: 1760
Description of the goods	: CORROSIVE LIQUID, N.O.S. (acetic acid, Nitric acid)
Class	: 8
Packing group	: II
Marine pollutant	: no

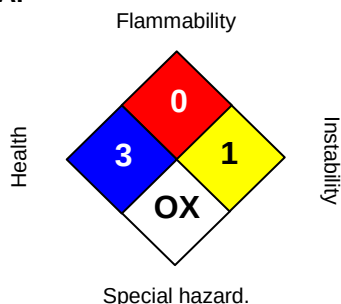
SECTION 15. REGULATORY INFORMATION

Mexico: Our MSDS meets the requirements established by the Official Mexican Regulation NOM-018-STPS-2015, Harmonized System for the identification and communication of hazards and risks by dangerous chemical substances in work centers.

SECTION 16. OTHER INFORMATION

Product AS SOLD

NFPA:



HMIS III:

HEALTH	3
FLAMMABILITY	0
PHYSICAL HAZARD	1

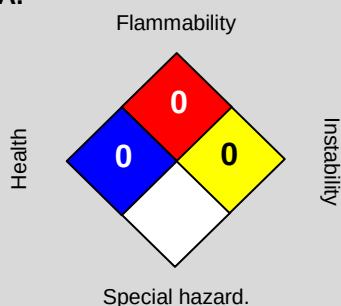
0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

SAFETY DATA SHEET

MANDATE PLUS

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 29.08.2017
Version : 1.0
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is considered correct but is not thorough and is intended for guidance only, which is based on current knowledge of the substance or mixture and is applicable to the appropriate safety precautions for the product.